

Agile trends in Chinese global software development industry: Fuzzy AHP based conceptual mapping

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ABSTRACT

Global Software Development (GSD) has gained great attention during the past decade or so, and it is also being adopted by most of the software development organizations in China and a wide range of firms have established its operations in various Chinese cities. However, GSD practices becomes more challenging for the projects based on the agile concepts. The aim of this study is to develop a taxonomy of the factors that could positively impact the scaling process of agile methods in the Chinese GSD industry. The factors are identified by exploring the available literature and conducting industrial study with the Chinese agile and GSD practitioners. The finally reported factors are categorized, prioritized and developed their taxonomy using the multi-criterion decision making (MCDM) Fuzzy AHP approach. The given taxonomy is significant and progressive for GSD industry to assess and improve the scaling process of agile methods.

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1. Introduction

Global software development (GSD) is a software development paradigm, where the team members are geographically, temporally and culturally dispersed [1]. They virtually coordinate, communicate and perform the development activities [1]. The geographically distributed software development concept was presented quite earlier than the word GSD. According to Lee et al. [2], in 1970's the "contract programming" method was introduced, where the software development organizations outsourced the system components to the third-party for development. Majority of the organizations in west (client) start outsourcing to vendor firms in the east because of various issues including cost, lack of resources, time and skilled workforce [3,4].

Economic and strategic benefits are the key factors that motivated the software industry towards globalization [5–7]. The

survey report published by the Standish Group revealed that 20% client (western) organizations outsource the software projects to vendor (eastern) firms across the world [8]. The main reason of the increase in offshore software outsourcing (global software development) are the potential benefits offered by the vendor organization, including low cost, 24/7 development, access to the international market, high-quality product and skilled human resources [6,9].

Presently, the concept of GSD is growing fast in China and India [10]. Majority of the multinational software development enterprises are outsourcing the projects to Chinese software industry [10–12]. There are twenty-one cities in China that are considered the model cities for GSD, while other cities are also struggling to establish the GSD industry [13]. Software development companies that are seeking growth could benefit from outsourcing their projects to China for number of reasons i.e. economic incentives, synergy, developed infrastructure and government support [13,14].

However, adopting GSD practices in the Chinese software industry is not a straightforward process and the development teams face various challenges due to geographical, socio-cultural and temporal distances [14]. The challenges included the tight project deadlines, business and market needs, system quality, customer satisfaction and frequent feedback [5,15]. The given challenges motivated different Chinese software development enterprises to adopt agile methods in the domain of GSD [16,

Abbreviations: Global Software Development, GSD; Multi-Criterion Decision Making, MCDM; Scaled Agile Framework, SAgile; Agile Process Maturity Model, APMM; Disciplined Agile Delivery, DAD; Large-Scale Scrum, LeSS; Extreme Programming, XP; Feature Driven Development, FDD; Agile Data, AD; Unified Process, UP; Dynamic Systems Development Method, DSDM; Fuzzy Analytic Hierarchy Process, AHP; Local Weight, LW; Global Weight, GW; Local Rank, LR; Global Rank, GR

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[17]. The integration of agile and GSD assist the firms to get the speed and quality benefits of agile methods along with the cost benefits of GSD [7]. Agile software development can be defined as a lightweight software development framework that follows an iterative and incremental process to support flexible, rapid, and effective development under time pressure, change and uncertainties [18]. The key principles of agile manifesto include customer collaboration, communication, instant feedback, teamwork, fast product delivery, simplicity and respond to requirement changes [18–20]. The agile manifesto was designed in 2001 by the group of system and software development experts based on their knowledge and understanding of agile concepts [18]. The most common and widely used agile frameworks include SCRUM, extreme programming (XP), Kanban, lean, feature driven development (FDD) and dynamic systems development method (DSDM) [19].

Despite the advantages of agile development, scaling agile methods in GSD environment brings additional challenges and associated risks which make the development activities more puzzling [7]. Paasivaara and Lassenius [21] mentioned that GSD need formal communication and coordination protocols for the team members located across different geographical locations, on the other hand, the agile methods are entirely based on small-sized collocated (single site) teams of software professionals and customers in an intense collaborative environment [22]. Similarly, in China, the cultural norms, values and language barriers make the scaling process of agile practices more challenging [17]. The chance of miscommunication, lack of coordination and control greatly increased because of language and cultural differences. Language is one of the first communication barriers [23]. Mandarin is the national and official language in China and most of the clients working with Chinese vendors speak either English as a first language or primary language [16,23]. The second important communication barrier is the communication style. Majority of the western clients communicate quite directly and in a concise way, on the other hand in China, the people convey the message by using different analogies and stories [17]. One could understand the message by filtering the stories and uncover the important statement. The third critical barrier is the organizational culture. For example, it is acceptable in western society to say “no” or response negative to top management if they ask about something [23]. There could be issues which arise and prevent the team members from completing the task on time and it is perfectly fine to let the top management know about those issues. However, it is not that straightforward in Chinese organizational culture and considers rude to say no to the boss or superiors. It could also be considered irresponsible behavior in case of not completing the task on time [16,17]. Based on the above discussion, it becomes much demanding to manage the agile activities in the Chinese GSD industry, as agile concepts are merely based on communication and close coordination. Presently, China is becoming a leading software outsourcing hub [24] and it is critical to tackle the key issues faced by the agile teams while they perform the software development activities in GSD environment and communicate with client organizations across the globe.

2. Research motivation

Various models and frameworks are designed to scale agile methods and assess the organizational agility capabilities i.e. Scaled Agile Framework (SAFe) [25], Agile Process Maturity Model (APMM) [26] and Disciplined Agile Delivery (DAD) [27]. Paasivaara et al. [22] reported that these models and standards have been successfully implemented in large scale organizations, but they are not explicitly addressing the distributed nature of

GSD projects [28,29]. Paasivaara [30] conducted an industrial case study and reported the scaling experience of SAFe framework in a globally distributed firm. The results of the case study revealed different challenges of using SAFe framework in GSD environment, including change resistance, engaging people, SAFe training, introducing program increment planning etc [31]. Similarly, IBM developed the Agile Process Maturity Model (APMM) to provide guidance for adopting agile techniques in a specific domain [26], however, the proposed model did not explore the geographically distributed nature of software development. The model has not focused on specific strategies for scaling agile practices in GSD environment. Moreover, hybrid DAD framework was developed that adopt and integrate the proven practices of agile methods including Kanban, extreme programming (XP), agile data (AD) and unified process (UP). It allows an organization to customize the agile practices based on the project needs and demands. However, the agile frameworks comparative study conducted by Ebert and Paasivaara [32] exhibit that DAD is a feasible choice regarding project scope and cost, but it is a challenging option for globally distributed teams. Moreover, Paasivaara and Lassenius [21] conducted a case study of agile transformation in a large-scale organization ‘Comptel’. The authors reported the case study findings by highlighting the significance of agile methods in globally distributed environment.

The deployment process of agile methods is more hard in GSD environment as compared to collocated (single site) [29,33–37]. More specifically, state-of-the-art research have not focussed on developing a strategy that could be used to manage agile methods in the domain of Chinese GSD industry [16,17,38,39]. The cultural and geographically diverse nature of the Chinese industry make the agile scaling process more challenging [16,17]. The cultural differences could negatively impact the collaborative work setting and organizational relationship, which are the key attributes of agile methods [17]. The current software outsourcing landscape in China motivate this work to explore the significance of agile methods that every client and vendor organization should think about.

Based on the given discussion, this study is conducted with the aim to develop a taxonomy of the factors that could positively impact the agile scaling process in Chinese GSD industry. The taxonomy is based on the agile success factors identified during the literature survey and industrial study conducted with the practitioners. The key objective of the industrial study is to know the perceptions and opinions of agile practitioners working in Chinese GSD industry. However, it is challenging to prioritize multiple factors based on the practitioner’s opinions that could bring vagueness and uncertainties. Quantitative prediction is challenging for humans, as they could more perfectly convey the feelings verbally (qualitatively). Therefore, we used the Fuzzy AHP approach to translate the qualitative prediction of the agile and GSD practitioners into quantitative prioritization values. It is a well-known approach that usually use for rating the human based multi criteria decision making problems. Fuzzy AHP approach has previously been used in different other studies.

For example, Singh and Prasher [40] evaluate the quality of services in different hospitals and prioritize the healthcare service quality attributes using Fuzzy AHP approach. The prioritization was eventually used for listing the best hospitals based on the quality of their services. Similarly, Wang et al. [41] used Fuzzy AHP to select the most common sustainability problems for both society and business in order to develop a framework for management and strategic planning. They mentioned that the framework works as a decision-making tool for organizational management while they work on sustainability related issues. Moreover, Yucesan and Kahraman [42] identified, categorized and prioritize various safety and financial risks in a hydroelectric

plant. Fuzzy AHP approach is used to list down the risks based on their prioritization and significance. Additionally, best practices were provided to mitigate the identified risks, maintain the safety and overcome the financial losses.

Therefore, the use of Fuzzy AHP approach in the above most recent studies motivated us to follow its concepts and develop the taxonomy of agile factors and their categories. The taxonomy provide a robust framework for organizations who are interested to scale the agile methods in GSD environment. Following are the research questions developed to achieve the key objectives:

RQ1. What are the success factors of scaling agile methods in the domain of Chinese GSD industry?

RQ2. How a practically robust taxonomy of the factors could be developed based on their prioritization?

3. Research methodology

The data are collected and analyzed based on the literature survey and industrial empirical study conducted with the agile practitioners working in Chinese GSD industry.

3.1. Literature survey

The literature survey is conducted to identify the key factors that could positively impact the scaling process of agile methods in GSD environment. The snowballing data sampling approach is used to conduct the literature survey, which is based on the concept of selecting the literature studies found in the reference section (backward snowballing) of the relevant publications as well the studies citing (forward snowballing) the selected studies [43]. The sample size of the selected studies steadily increases as more references and citations are explored [43]. The relevant literature studies are listed in order to extract the factors that could be significant in the scaling process of agile methods. Moreover, those studies are also considered, where the factors are not explicitly discussed [44,45], but presented the relevant agile lesson learned and experience reports. Identifying and extracting factors from such reports are more challenging because it required complete and in-depth review [45]. The literature studies are searched using the Google Scholar search engine. It provides a simple interface to broadly search the scholarly articles available on different other common digital libraries e.g., Springer Link, IEEE Xplore, ACM Digital Library etc. It gives us confidence that no relevant digital library has been missed. The Google Scholar search engine is explored using the keywords of the study and identify the relevant published articles. The studies selection process is mainly performed by the first and second authors. However, the disagreements between both authors at any point have been settled based on the discussion and overview of all the authors. We finally shortlisted 92 studies (references list) using both forward and back snowballing. The studies are considered to structure this article, as well address the research questions discussed in Section 1.

The first and second authors reviewed the selected studies and develop the list of the success factors that could positively impact the scaling process of agile methods. The second review of the studies is done by the third and fourth authors in order to refine the results of the first review and report the missing information. Moreover, two external reviewers are invited to evaluate the interpersonal biases in the review process. The external reviewers are requested to randomly select 15 articles and conduct the review process as performed by the authors. The interpersonal biases between the external reviewers and the authors have been assessed by performing the Kendalls coefficient of concordance (W) test. Kendalls coefficient of concordance (W) is a well-known

Table 1

Kendalls coefficient of concordance (W) test values.

Data Set	Chi-Square	df	Total variables	Evaluators	Probability value	W
Agile	34.733	14	15	3	0.001	0.826

statistical approach used to identify the level of agreement between a group of people that evaluate a set consist of n objects [46]. The range of the W assessment score is from 0 to 1, where $W=0$ is showing complete disagreement level between the people and $W=1$ refer to complete agreement [46]. In this study, the code for Kendalls coefficient of concordance (W) test is developed using R-3.6.3 and calculated the W value to assess the level of agreement for the literature studies review process between the authors and the external reviewers. The results given in Table 1 ($W=0.826$) shows that the invited external reviewers and the authors are at positive agreement level for the studies selection and data extraction process. The following is the R code for the Kendalls coefficient of concordance (W):

```
library(DescTools)
Agile<-data.frame(
external_rev1 = c(6,5,7,1,10,4,8,3,9,2,12,11,15,13,14),
external_rev2 = c(6,7,5,2,1,4,10,9,8,3,11,14,13,15,12),
authors = c(4,5,7,1,2,3,15,9,8,6,14,10,12,11,13))
KendallW(Agile, TRUE)
)
KendallW(Agile, TRUE, test=TRUE)
)
KendallW(t(d.att[, -1]), test = TRUE)
```

3.2. Industrial empirical study

The survey data collection approach is used to validate the literature findings and understand the perceptions of the Chinese GSD industry practitioners regarding the significance of the agile scaling process. Survey approach assists to collect data from a large group of people (targeted population) [47]. It could help to extract data related to people attitudes, that are otherwise challenging using the observational methods [47]. Following are the key steps of conducting the industrial survey study.

3.2.1. Survey questionnaire development and pilot evaluation

It is important to empirically evaluate the findings of the literature study and know the industrial understanding of those findings. Different factors that could impact the scaling process of agile methods in GSD environment are identified during the literature survey (Section 4). To make the identified factors more specific in the domain of Chinese GSD industry, we developed a survey questionnaire to report the views and opinions of the practitioners.

The survey questionnaire consists of both open and close-ended questions. The close-ended questions are developed based on the literature survey findings (i.e., success factors). The assessment scale of the close-ended questions consists of a five-point Likert scale ("strongly agree, agree, neutral, disagree, and strongly disagree"). The selection of the five-point Likert scale removes the biases in the survey responses by providing the neutral option [1,48]. The absence of neutral option could actually force the survey participants to select either negative or positive option even if they have a neutral opinion about a specific variable [1,48]. Similarly, the open-ended questions are provided to report any new related factors, that are not discussed in the literature studies.

The structure of the survey instrument is evaluated to improve the understandability and readability of the questionnaire variables. Three qualitative software engineering research experts are invited to provide feedback about the survey questionnaire based on their experience and understanding of the survey questions. The experts are affiliated with the Wuhan University, China, Nanjing University, China and Molde University College (Specialized University in Logistics), Norway. They are requested to fill the questionnaire and provide their suggestions and recommendations related to the overall design and understandability of the questionnaire variables. The questionnaire is updated based on their recommendations, that are mostly related to the structure of the survey instrument. They suggested minor changes in the design related to the clarity of the questionnaire variables. Moreover, two of the experts recommended to present the variables across the five-Likert scale rather than defining them as separate questions that might make difficult to interpret the actual meaning of the variables.

3.2.2. Data sources of the industrial study

The research domain for this study is the Chinese global software development industry. We used the simple random sampling approach to collect data from multiple organizations [49]. Simple random sampling provides an opportunity to collect data from the large targeted population located across the geographical boundaries [47]. We were not able to visit and collect the data from all the organizations of the targeted population across China, therefore the random sampling approach makes it easy by giving an equal selection chance to each member of the population.

In Nanjing city (China), the targeted population for the survey study are physically approached by taking permission from the concerned executive working in Nanjing New and High Technology Industry Development Zone (Nanjing Jiangbei New Area). We requested the park executive and she forward our request to different software development organizations working in the domain of agile and GSD. Eventually, ten organizations agreed to let the concern practitioners participate in our survey. The first and fourth author visited the agreed organizations in the Nanjing science park and collect data from a total of 35 participants.

The organizations working outside Nanjing City are contacted online by email and professional social media networks i.e., WeChat, QQ, LinkedIn and ResearchGate. Finally, 48 practitioners participated in the survey and provide their responses. Therefore, a total of 83 responses (Nanjing+ other cities) are considered for the final data analysis. The data collection process was started in November 2019 and finished in January 2020. The survey questionnaire sample is provided in [Appendix A](#).

3.2.3. Analysis of the empirical data

The data collected from the 83 survey participants are analyzed using the frequency analysis method. The survey data is of descriptive nature and frequency analysis is considered the best approach for descriptive data [50]. It comparatively analyzed the data variables and compute the agreement level between the survey participants based on the selected Likert scale. Frequency analysis method has also been adopted in different other software engineering empirical studies [51,52].

4. Literature survey findings

The literature studies are investigated based on the criteria discussed in Section 3.1 and identified a total of 23 variables that could be success factors (SF) for scaling agile methods in the domain of GSD industry. The list of the identified success factors is given in [Table 2](#) and discussed hereafter.

Leadership support (SF1): In distributed work sites, it is important that one member of the team of three or more should serve as

the team lead. The team lead could be a senior developer, project manager, scrum master etc., who has potential team building abilities [53]. The team lead at work site will provide all the assistance and project directions to the team members. They will streamline the communication with other distributed sites and ensure that all the project updates are properly circulated in order to stay on track [53,54].

Knowledge acquisition (SF2): Proper coaching and training are required on agile practices and processes in order to scale the agile methods in a globally distributed environment [55]. Teams usually adopt non-agile practices because of lack of knowledge, training and coaching [56]. It is important to keep engaged the agile team members in external learning events and provide knowledge and understanding by conducting training sessions and workshops.

Engineering excellence (SF3): Strong engineering culture is important for scaling agile methods. Improper engineering practices could significantly impact team productivity and results in the delivery of low quality product [57]. The continuous product delivery demand could cause technical debts that need more team effort and the product quality might suffer [57]. GSD organizations should make sure engineering excellence while scaling agile methods by deploying test automation, building DevOps culture, continuous collaboration across the distributed sites and use the visual environment to track the product progress and quality [57,58].

Tools and infrastructure (SF4): GSD organizations need to invest in modern tools and infrastructure for smooth agile scaling process. Complexity in the software development process increases when the agile teams perform the development activities across the geographical boundaries [59]. The modern-day software engineering tools and infrastructures such as continuous integration, test automation and cloud native platform are the key components of the agile scaling process. It is very important to use the most advance tools and techniques in order to support the deployment of agile practices across geographically distributed environment [60].

Communities of practice (SF5): It is a group of people who have a common interest related to a specific business domain [56,57]. They regularly collaborate and share information and knowledge to tackle key challenges in that specific domain. Communities of practice are common for continues delivery, DevOps, scrum masters and agile testing [61]. The key individuals in communities of practice provide frameworks and guidance to maintain consistency across the agile teams [56].

Agile champions and change agents (SF6): Agile champions and agents are the key catalysts that drive change between the management and teams [59]. They are important to scale the agile methods in a large-scale environment and develop the change culture in the organization. There should be concrete understanding and well thought decisions for scaling agile in GSD environment [61]. The agile scaling could create ripples that might not go well with some of the stakeholders, therefore it is the job of agile champions and agents to buy-in the top management support.

Arrange social activities (SF7): Social activities help to build the scaling agile mindset. Scaling agile could be driven by social activities and events where the teams and management receive and share the information to find new ways of working [62,63]. Agile engaging activities could increase awareness and motivate the teams in adopting agile practices. Social activities work as a tool for building bonds between the teams, management and customers [57,64].

Table 2
Identified success factors.

S. No	Success Factor	References
SF1	Leadership support	[53,54]
SF2	Knowledge acquisition	[55,56]
SF3	Engineering excellence	[57,58]
SF4	Tools and infrastructure	[59,60]
SF5	Communities of practices	[56,57,61]
SF6	Agile champions and change agents	[59,61]
SF7	Arrange social activities	[57,62,63,65]
SF8	Customer involvement	[62,63]
SF9	Creating self-organizing teams	[18,59,66]
SF10	Selecting and customizing the agile approach	[59]
SF11	Workflow transparency	[67–70]
SF12	Training and coaching	[58,69]
SF13	Communication and control	[58,67,70,71,71]
SF14	Organizational vision	[58,66]
SF15	Continuous learning environment	[65]
SF16	Instant feedback	[70,72]
SF17	Skilled human resources	[56,62]
SF18	Start with minimum viable product (MVP)	[59,73,74]
SF19	Create a single product backlog	[63,75,76]
SF20	Build a collaborative culture	[57,70,77]
SF21	Use a large-scale agile framework	[58,65,75,78]
SF22	Planning for scaling process	[58,63,79,80]
SF23	Maturity model for agile transformation	[31,81]

Customer Involvement (SF8): Customer involvement significantly influence the scaling process of agile methods [62]. Active customer collaboration and risk tolerance could increase the team ability of agile transformation [63]. The customer should not be placed on the audience seat and put the burden of scaling on the agile team [63]. It could only be successful when both the customer and teams together appear at the center stage.

Creating self-organizing teams (SF9): One of the key principles of agile manifesto defines that the best product design, requirements and architecture emerge from the self-organizing teams [18]. Self-organizing are the teams where the team members manage the key responsibilities and project timelines without external influence [59]. They are managed by the team manager and they internally decide the roles and responsibilities. The key principles to develop the self-organizing team include trust, collaboration, responsibility, motivation, teamwork and ownership [66]. The concept of self-organizing teams increases the continuous delivery, customer focus, quality and satisfaction.

Selecting and customizing the agile approach (SF10): Different agile methods work in different circumstances. Each agile team has unique characteristics and they adopt specific agile approach [59]. Off the shelf approaches, e.g., XP, Scrum and Kanban could use by some teams but it is not the case for all. Therefore, it is important to customize the agile approaches based on the organizational and team circumstances, which might require specific planning and arrangements [59].

Workflow transparency (SF11): Work transparency in the agile environment provides a deep insight into the project activities [67,68]. It makes easier to track the project progress, including ongoing tasks, assignments in the queue, time frame and any roadblocks obstructing agile teams while performing specific activities [69,70]. Work transparency could assist to effectively manage the agile teams working beyond the geographical boundaries.

Training and coaching (SF12): Scaling agile methods in GSD environment bring significant change in an organizational setting [69]. Training and coaching improve the scaling capabilities by motivating the agile teams and building agile mindset. It makes people more positive and gives an understanding of the new

working environment and culture [58]. Lack of training and coaching could undermine the scaling process by improperly deploying agile practices.

Communication and control (SF13): Various communication and control issues arise while scaling agile in a distributed environment [67,71]. Effective communication in a large scale distributed environment comprised of effective strategies that contribute to communication and control activities [58]. Structure, synchronization and boundary spanning are the three key components that significantly impact the coordination and communication effectiveness [70,71].

Organizational vision (SF14): The agile practices either driven from the top-down or bottom-up direction. It is important that organizational management involves in both scenarios of agile transformation and continuously provide training or education opportunities at all levels including project managers, executives, and agile team members [58,66]. Continuous support and transparency should include at both top and bottom levels.

Continuous learning environment (SF15): Continuous learning provides an opportunity to achieve creativity, experience and increase knowledge of organizational workforce and customers [64]. Organization could develop the learning environment for agile transformation by considering the core three dimensions i.e.

- Team members at the top and bottom level should learn in order to transform the organization towards the world of continuous changing.
- Organizational management should encourage the employees to explore and create novel ideas for future value delivery.
- Enterprise should focus on continuous improvement of agile practices and transformation techniques.

Instant feedback (SF16): Agile is an incremental and collaborative approach to satisfy the product customers and stakeholders through continuous delivery of a functional software system [70]. To make the agile process more collaborative and continuous, it is important that the development and operation teams focus on short and frequent feedback loops [70]. Frequent feedback across the distributed agile teams and customers will motivate the practitioners to focus on the most relevant system features. Moreover, it will make easy the change management process, specifically, when the new requirements arise or demanded by the customers [72].

Skilled human resources (SF17): Skilled human resources is a critical factor for scaling the agile methods in GSD environment. The agile team members and top management should have professional certification and experience in agile or relevant fields [56]. They should have the expertise of agile and GSD projects, which could give them an understanding of challenges faced during the agile scaling process. The skilled practitioners could tackle the issues that arise during the agile scaling process because of geographical, cultural and temporal boundaries in GSD [62].

Start with the minimum viable product (MVP) (SF18): MVP is a concept that is used to get an understanding of the new product impact [73]. It allows the agile teams to get maximum learning about a new product based on customer feedback [59,74]. It is an iterative process that facilitates the product development activities by bridging the communication gap between the business stakeholders and the agile team [73].

Create a single product backlog (SF19): Backlog consists of key features and scope which should consider in the product [75]. Backlog elements are categorized and prioritized based on their significance level [63,76]. Each element of the backlog is tagged

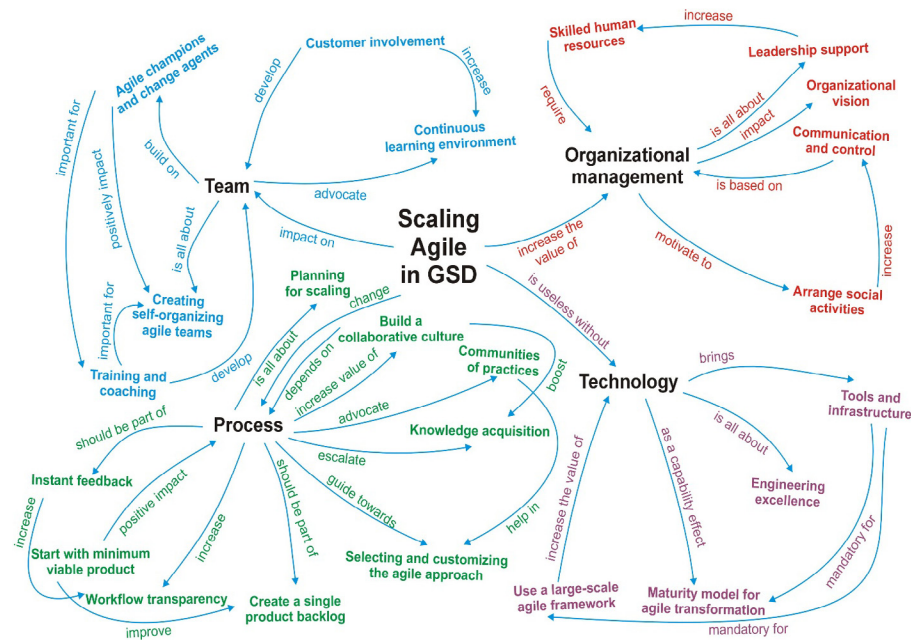


Fig. 1. Conceptual mapping of the identified success factors.

with a specific serial number. Single product backlog is a critical factor when the agile teams perform the development activities across distributed sites. It will give a single overview of all the product elements and features to the agile team and product customers [76].

Build a collaborative culture (SF20): Customer collaboration over contract negotiation is one of the listed agile manifesto values [57]. Collaboration in an agile environment occurs bidirectionally between the customer and the agile team. Collaboration culture could be built by understanding the team composition, developing the user-focused environment, integrating continuous monitoring and feedback, identifying the key barriers and choosing the right collaboration tool for the distributed teams [70,77].

Use a large-scale agile framework (SF21): There are multiple agile scale frameworks e.g. Scaled Agile Framework (SAFe) [25], Large Scale Scrum (LeSS) [82], and Disciplined Agile Delivery (DAD) [27]. The most important concern is to select a suitable framework that could fit the organization's needs [65]. All the available frameworks have pros and cons based on the type of organization. There is not a "one size fits all" approach, however, it is very important to choose the appropriate framework that returns on investment in the form of performance, improves cross-team communication, delivers high-value end product and adjusts priorities at low cost [58,75,78].

Planning for scaling process (SF22): Agile scaling in a distributed environment is a challenging process and it will not be complete within days. The end goal could only be achieved if the organization effectively plans the scaling strategy [79,80]. Regardless of using any framework, following are the key indicators that should be part of agile scaling planning: define roles and responsibilities, improve dependency, balance top-down and bottom-up coordination, embracing customer-centric development and adoption of lean practices [58,63].

Maturity model for agile transformation (SF23): Agile scalability and integration in traditional GSD environment is still considered a challenging concern. Existing frameworks, i.e. Scaled Agile Framework (SAFe) [25], Large-scale Scrum (LeSS) [82] and Disciplined Agile Delivery (DAD) [27] etc. developed for large

scale agile transformation but there is a need of specific well-structured model that could manage the agile practices in GSD environment [31,81]. Large scale and GSD are two different domains and it is not the case that large scale organization should be globally distributed. This issue could only be addressed by using a specific agile transformation maturity model that assesses the scaling capabilities of a GSD organization and provides the best improvement practices [31].

5. Conceptual mapping of the identified factors

The concepts of the Ramasubbu [5] process improvement framework have been used to conceptually classify the identified factors across four main categories i.e. ("organizational management, team, technology and process"). The Ramasubbu [5] framework is selected because both studies focus on process improvement activities in the domain of GSD. The factors categorization is informally performed and presented as the conceptual map (Fig. 1, Table 3). The mapping process is conducted by the authors of the study based on the literature findings and understanding of GSD and agile concepts.

Further assessment has been done by sharing the conceptual map and categorization with the global software engineering research experts working with the Norwegian University of Science and Technology (NTNU) and Molde University College (Specialized University in Logistics), Norway. The invited experts suggested some changes regarding the positions and relationships of the factors. For example, one of the experts recommended to move the success factor *training and coaching* (SF12) from process category to the team. He mentioned that training and coaching are more people and team-oriented activities and is best suited to consider in the team category. We finally made the suggested changes and revised the factors conceptual mapping (Fig. 1, Table 3).

The results given in Table 3 and Fig. 1 show that *leadership support* (SF1), *communication and control* (SF7), *arrange social activities* (SF13), *organizational vision* (SF14) and *skilled human resources* (SF17) are related to the organizational management category of the conceptual map. Similarly, *customer involvement* (SF8), *agile champions and change agents* (SF6), *creating self-organizing*

Table 3
Categorical classification of the identified success factors.

Category	Factors
Organizational Management	Leadership support (SF1) Arrange social activities (SF7) Communication and control (SF13) Organizational vision (SF14) Skilled human resources (SF17)
Team	Customer involvement (SF8) Agile champions and change agents (SF6) Creating self-organizing agile teams (SF9) Training and coaching (SF12) Continuous learning environment (SF15)
Technology	Engineering excellence (SF3) Tools and infrastructure (SF4) Maturity model for agile transformation (SF23) Use a large-scale agile framework (SF21)
Process	Communities of practices (SF5) Knowledge acquisition (SF2) Selecting and customizing the agile approach (SF10) Workflow transparency (SF11) Start with minimum viable product (MVP) (SF18) Instant feedback (SF16) Build a collaborative culture (SF20) Create a single product backlog (SF19) Planning for scaling (SF22)

agile teams (SF9), training and coaching (SF12) and continuous learning environment (SF15) are in line with the team category. Engineering excellence (SF3), tools and infrastructure (SF4), maturity model for agile transformation (SF23) and use a large-scale agile framework (SF21) fell within the technology category. Moreover, Communities of practices (SF5), knowledge acquisition (SF2), selecting and customizing the agile approach (SF10), workflow transparency (SF11), start with minimum viable product (MVP) (SF18), instant feedback (SF16), build a collaborative culture (SF20), create a single product backlog (SF19), and planning for scaling (SF22) are associated with the process category.

The conceptual classification provides a theoretical framework and understanding of success factors that are significant for scaling agile methods in GSD environment. It offers a roadmap to follow before initiating the scaling process.

6. Industrial empirical evaluation of literature findings

The findings of the literature study consist of total 23 success factors (Section 4) that are further mapped into four core categories and presented as the conceptual map (Section 5). However, it is significant to empirically validate the literature findings and the conceptual mapping process. The empirical assessment is conducted using the survey questionnaire approach. The survey instrument is used to collect the data from the practitioners working on agile and GSD projects in the Chinese software industry. The aim of the survey study is to know the understating and perceptions of the practitioners working in Chinese GSD industry and their feedback regarding the identified factors and their conceptual mapping. The survey study data are collected from a total of 83 participants and the details of the data collection process and data sampling are thoroughly discussed in Section 3.2. Moreover, Appendix B consists of the survey respondents demographic details.

The data collected using the five-point Likert scale are classified across positive, negative and neutral categories (Table 4). The positive category provides the details of the survey participants who positively agree with the results of the literature survey and the conceptual mapping. They show positive agreement with the literature findings by considering the identified factors for the agile scaling process in the domain of Chinese GSD industry. Similarly, the survey respondents who do not agree to consider

the identified factors for agile scaling process are reported in the negative category. The neutral category provides the details of those participants who are not sure about the significance of the factors and their mapping. The results given in Table 4 revealed that the reported factors and their categorization have been positively considered by the majority (>70%) of the survey participants.

Communication and control (SF13, 95%) is the most important factor in the positive category and it is reported by 95% of the respondents. The basic concepts of agile software development are based on close communication, coordination and control [18]. Compared to collocated (single-site) software development, communication in GSD environment is affected by the cultural, temporal and geographical boundaries [83]. Cultural factors and most importantly the language differences may cause miscommunication and lack of control [17]. Language seems to be the major challenge for Chinese companies to work on agile projects with client organizations. English is common and official language in most of the countries, on the other hand, Mandarin is the national and widely spoken language in China region [16]. Communication and collaboration are considered the key parameters of the agile software development manifesto [18], which becomes much challenging in GSD environment. The above discussion justify the high response rate (95%) of the survey participants for the communication and control (SF13) factor. They have a strong understanding of the importance of communication and control activities for agile projects in the domain of GSD.

Similarly, use a large-scale agile framework (SF21, 23%) is considered the most common factor in the negative category. There are different frameworks developed to scale the agile methods in the large-scale software development environment e.g., Scaled Agile Framework (SAFe), and Large-Scale Scrum (LeSS). These frameworks provide key principles, a set of experiments and guideline to scale the agile methods for large scale projects, however, they are not able to manage the agile activities in the domain of GSD environment (Dikert, [21,22,29]. They did not recommend any implementation strategies to specifically mitigate GSD challenges. Moreover, these models introduce complex management principles which are not in line with the agile lightweight practices. Arrange social activities (SF7, 19%) is viewed as the most neutral factor by the survey participants. The physical

Table 4
Frequency analysis of industrial survey data.

Identified factors and Categories	Sample size (n = 83)							
	Positive			Negative			Neutral	
	S.A	A	%	S.D	D	%	N	%
Organizational Management	40	35	90	0	2	2	6	7
Leadership support (SF1)	28	39	81	0	8	10	8	10
Arrange social activities (SF7)	30	33	76	2	2	5	16	19
Communication and control (SF13)	42	37	95	0	0	0	4	5
Organizational vision (SF14)	31	29	72	5	5	12	13	16
Skilled human resources (SF17)	36	32	82	2	3	6	10	12
Team	38	31	83	1	1	2	12	14
Customer involvement (SF8)	29	30	71	3	6	11	15	18
Agile champions and change agents (SF6)	34	40	89	0	1	1	8	11
Creating self-organizing agile teams (SF9)	27	35	75	3	5	10	13	16
Training and coaching (SF12)	31	33	77	0	4	5	15	18
Continuous learning environment (SF15)	38	24	75	7	7	17	7	8
Technology	44	28	87	0	0	0	11	13
Engineering excellence (SF3)	36	34	84	2	2	5	9	11
Tools and infrastructure (SF4)	30	30	72	5	4	11	14	17
Maturity model for agile transformation (SF23)	41	37	93	0	0	0	5	6
Use a large-scale agile framework (SF21)	32	28	72	9	10	23	4	5
Process	37	33	81	2	5	8	9	11
Communities of practices (SF5)	29	30	71	4	8	14	12	14
Knowledge acquisition (SF2)	32	31	76	4	6	12	10	12
Selecting and customizing the agile approach (SF10)	38	33	86	1	3	5	8	11
Workflow transparency (SF11)	43	29	87	3	5	10	3	4
Start with minimum viable product (MVP) (SF18)	27	36	76	4	9	16	7	8
Instant feedback (SF16)	40	27	81	0	3	4	13	16
Build a collaborative culture (SF20)	39	35	90	0	0	0	9	11
Create a single product backlog (SF19)	25	30	66	8	5	16	15	18
Planning for scaling (SF22)	37	33	84	0	2	2	11	13

Strongly agree: SA, Agree: A, Strongly disagree: SD, Disagree: D, Neutral: N.

and cultural gap in GSD setting makes it challenging to arrange and engage the distributed teams in social activities. The survey respondents are based in Chinese GSD industry and they might not sure about the experience of social engagement with team members from different cultural backgrounds.

Finally, the survey participants provided additional factors in the open-ended section of the questionnaire. However, those factors were not considered as new, because their actual meanings were replicated with the identified factors (Table 2)

7. Fuzzy set theory and analytic hierarchy process (AHP)

In this section, we introduce the Fuzzy set theory and AHP approach that is used to prioritize the identified factors and their respective categories based on the data collected from the 23 industrial practitioners. Zadeh et al. [84] introduced a Fuzzy set theory as an extension of the conventional AHP approach. It was developed to deal with the uncertainties and vagueness in the perceptions of decision-makers. It provides more consistent, accurate and precise judgment for both qualitative and quantitative multi-criteria problems. The Fuzzy set approach consists of membership function that maps the values of the object between 0 and 1 e.g., if the membership function value is 0 it means that the objects does not belong to the set, however, if the value is 1 then it shows a strong association with the set. The value of the membership function between 0 and 1 shows the partial association.

Moreover, the analytical hierarchy process (AHP) was introduced by [85] for multi-criteria decision-making problems based on the qualitative data collected from the decision-makers [85]. Following are the three main steps involve in the AHP approach.

- Step-1: The complex decision-making problem is presented as a hierarchy in the first step of AHP approach (Fig. 2). The hierarchical tree is used by the decision-makers to provide their preferences between the alternatives. The implication details of the hierarchical tree are provided in (step-1, Section 8).

- Step-2: Develop the pairwise comparison matrices in order to calculate the weight of each factor across the hierarchy levels.
- Step-3: Determine the final priority ranking based on the normalized weight calculated in the second step.

The simple AHP approach is not strong enough to manage the vagueness and uncertainties for multi-criteria decision making problems based on quantitative data. The integration of both Fuzzy set theory and AHP becomes a more significant approach to determine more specific and accurate judgments. It has the characteristics to manage decision-making problems based on both quantitative and qualitative data. The priority weight and prefer rating of the criteria is calculated using the triangular Fuzzy numbers. We followed the concepts of the Fuzzy AHP approach developed by Chang [86] to calculate the priority weight of the identified success factors and their respective categories.

Suppose, $V = \{v_1, v_2, \dots, v_n\}$ represent the set of the objects for main categories and $U = \{u_1, u_2, \dots, u_n\}$ is the goal set for each category. Chang [86] mentioned that the extensive analysis for each object and each goal (g_i) should be respectively performed. Therefore, using Eq. (1), the (m) extent analysis for each object is calculated.

$$X_{gi}^1, X_{gi}^2, \dots, X_{gi}^m, \quad i = 1, 2, \dots, n \quad (1)$$

where, X_{gi}^j ($j = 1, 2, \dots, m$) are triangular Fuzzy numbers (TFNs). The step-by-step extent analysis instructions provided by Chang [86] are as follow.

Step 1: The i th object Fuzzy synthetic extent value should calculate using Eq. (2).

$$S_i = \sum_{j=1}^m X_{gi}^j \otimes \left[\sum_{i=1}^n \sum_{j=1}^m X_{gi}^j \right]^{-1} \quad (2)$$

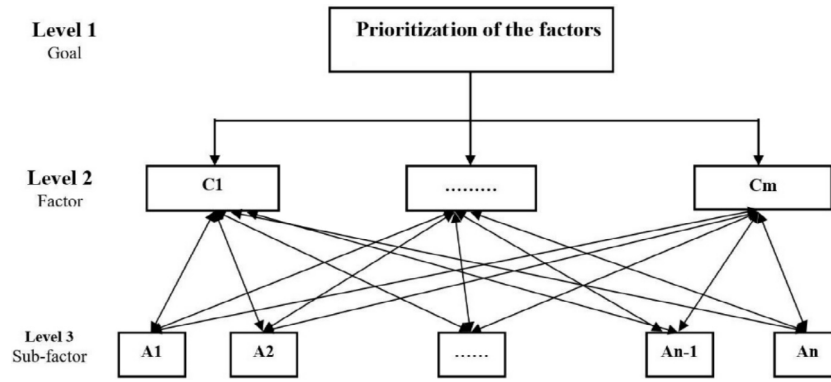


Fig. 2. AHP hierarchical tree.

where, the value of $\sum_{j=1}^m X_{gi}^j$ determine using the m extent analysis Fuzzy addition operation such as:

$$\sum_{j=1}^m X_{gi}^j = \left(\sum_{j=1}^m x_{gi}^l, \sum_{j=1}^m x_{gi}^m, \sum_{j=1}^m x_{gi}^r \right) \quad (3)$$

Similarly, the value of $\left[\sum_{i=1}^n \sum_{j=1}^m X_{gi}^j \right]^{-1}$ is achieved by executing the Fuzzy addition operation on X_{gi}^j ($j = 1, 2, \dots, m$) value:

$$\sum_{i=1}^n \sum_{j=1}^m X_{gi}^j = \left(\sum_{i=1}^n x_{gi}^l, \sum_{i=1}^n x_{gi}^m, \sum_{i=1}^n x_{gi}^r \right) \quad (4)$$

The Eq. (5) is used to calculate the inverse of the vector:

$$\left[\sum_{i=1}^n \sum_{j=1}^m X_{gi}^j \right]^{-1} = \left(\frac{1}{\sum_{i=1}^n x_{gi}^l}, \frac{1}{\sum_{i=1}^n x_{gi}^m}, \frac{1}{\sum_{i=1}^n x_{gi}^r} \right) \quad (5)$$

Step-2: Both X_1 and X_2 are triangular Fuzzy numbers and the Eq. (6) define the degree at which $X_2 = (x_2^l, x_2^m, x_2^r) \geq X_1 = (x_1^l, x_1^m, x_1^r)$

$$Y(X_2 \geq X_1) = \sup[\min(\mu_{x1}(v)), (\mu_{x2}(v))] \quad (6)$$

Eq. (6) can also be simplified as:

$$Y(X_2 \geq X_1) = \text{hgx}(X_1 \cap X_2) = \mu_{x2}(d) \\ = \begin{cases} 1 & \text{if } x_2^m \geq x_1^m \\ 0 & x_1^l \geq x_2^r \\ \frac{x_1^l x_2^r}{(x_2^m - x_2^r) - (x_1^m - x_1^l)} & \text{Otherwise} \end{cases} \quad (7)$$

where d is the highest intersection point between μ_{x1} and μ_{x2} (Fig. 3). The X_1 and X_2 could only be determined if we have the values of $Y_1 (X_2 \geq X_1)$ and $Y_2 (X_2 \geq X_1)$.

Step 3: The value of convex Fuzzy number X_i ($i = 1, 2, \dots, k$) could be calculated using the following equations.

$$Y(X \geq X_1, X_2, X_3, \dots, X_k) = \min Y(X \geq X_i) \quad (8)$$

suppose that,

$$d'(X_i) = \min Y(X_i \geq X_k) \quad (9)$$

for $k = 1, 2, \dots, n; k \neq i$.

The weight vector of each element could be calculated using Eq. (1).

$$W' = (d'(X_1), d'(X_2), d'(X_3), \dots, d'(X_n))^X \quad (10)$$

where, X_i ($i=1,2,\dots,n$) consist of n distinct elements.

Step 4: The priority weight criteria could be determined using Eq. (11), which normalize the weight vector and convert the results into a non-Fuzzy number.

$$W = (d(X_1), d(X_2), d(X_3), \dots, d(X_n))^X \quad (11)$$

The value of W represents a non-Fuzzy number.

Step 5: Checking consistency ratio: It is important to evaluate the consistency of the pairwise comparison matrices [85]. The graded mean integration approach could be used to measure the consistency ratio that converts the Fuzzy number of each matrix into the respective crisp value. Eq. (12) further use to convert the triangular Fuzzy number $P = (m, p, r)$ into the respective crisp number.

$$Q_{crisp} = \frac{(4m + p + r)}{6} \quad (12)$$

Once we calculate the crisp value (Q_{crisp}) then next use Eqs. (13) and (14) to determine the values of consistency index (CI) and consistency ratio (CR) respectively.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (13)$$

$$CR = \frac{CI}{RI} \quad (14)$$

where λ_{max} is the maximum eigenvalue, n is the size of the pairwise comparison matrix and the random index (RI) value could be find using Table 5.

The pairwise comparison matrices are consistent if the calculated value of $CR < 0.1$, otherwise the consistency ratio should evaluate again until the acceptable value of CR is achieved.

8. Success factors prioritization using fuzzy AHP approach

The step-by-step process of Fuzzy AHP approach discussed in Section 7 used to determine the priority weight of each identified success factor and respective categories. We use the ("MATLAB 13 on Intel core [TM] i5-4770") 495 platform to implement the Fuzzy AHP approach and following are the key results:

Step-1: In the first phase, we developed the hierarchal structure of the decision-making problem by classifying the key elements and their sub-elements across multiple levels [85]. It is significant to develop the hierarchal structure of the problem to minimum three levels, where the problem objective is defined at the level-1, the elements and sub-elements at level-2 and 3 respectively. The Figure-4 is the Fuzzy AHP hierarchal structure developed for the success factors and categories discussed in Section 4.5 respectively. The key goal of the study (i.e., multi-criterion decision-making model of success factors) is defined at the level-1 of

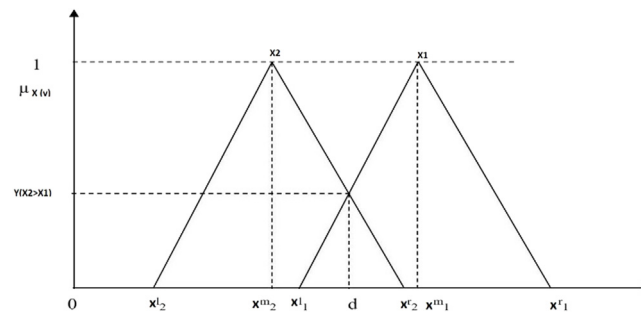


Fig. 3. Triangular Fuzzy number.

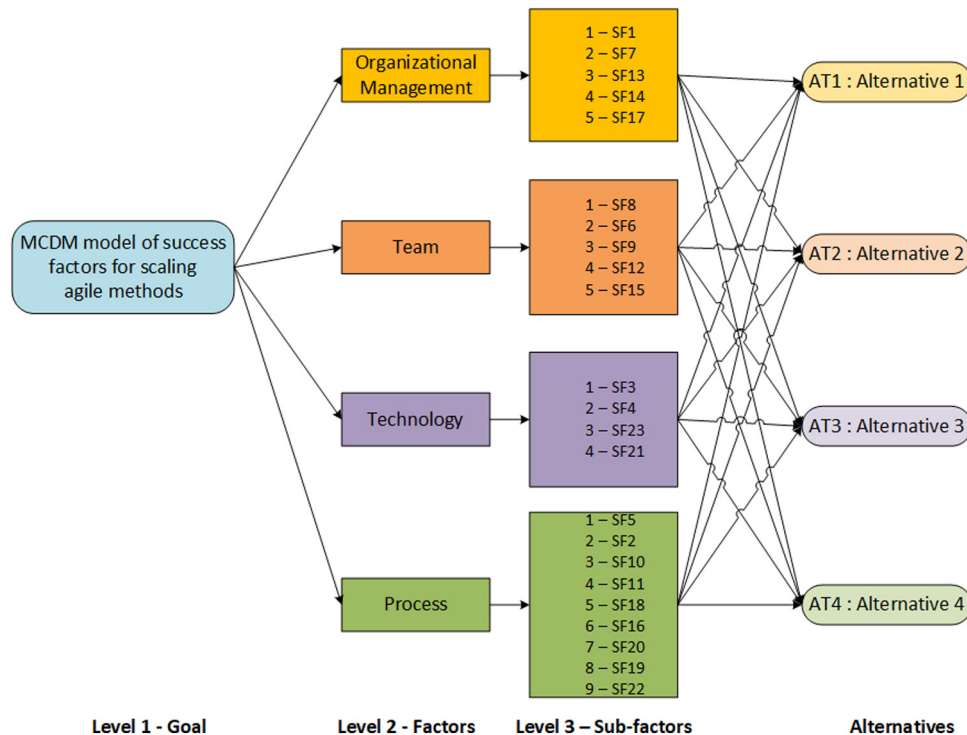


Fig. 4. The hierarchical structure of the decision-making problem.

hierarchical structure (Fig. 4). Similarly, the categories and the success factors are respectively defined at level-2 and level-3. The hierarchical structure provides an opportunity for the decision-makers to note their preference between the alternatives (Fig. 4). They provide a relative priority based on the TFN scale (Table 6).

Step-2: The second survey (Fuzzy AHP survey) is conducted to collect data from the experts and perform the pairwise comparison of the identified factors and the categories. The respondents of the first survey were invited to participate in the Fuzzy AHP survey. The first survey questionnaire was developed to know the opinions of the industrial practitioners regarding the literature findings and the conceptual mapping of the identified factors. The results are discussed in Section 6 and it revealed that majority of the survey participants are confidently sure to consider the reported factors for the scaling process of agile methods in the Chinese GSD industry.

The Fuzzy AHP survey questionnaire is developed based on the findings of the first survey study and criteria discussed in Section 3.2. The linguistic scale given in Table 6 is used to pairwise compare the reported factors and the categories. The sample Fuzzy AHP survey questionnaire is given in Appendix C.

The data are collected from 23 respondents, who shows positive agreement to provide their feedback for the prioritization

Table 5

RI value based on the matrix size.

Matrix Size (n)	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49

of the success factors and their categories based on the Chinese software industry. The generalization of the Fuzzy AHP survey results could be questionable due to the small size of the data sample. However, Fuzzy AHP is considered a subjective approach and best suited for small sample size [87,88]. For example, Yucesan and Kahraman [42] evaluated the operational risks of hydropower plants using Fuzzy AHP approach based on the data collected from five industrial experts. They identified total of twenty risk factors that could cause different hazards. In another study, Yildirim and Mercangoz [89] collected data from five experts and evaluated the logistic performance of Organization for Economic Co-operation and Development (OECD) member countries using Fuzzy AHP approach. They used the six key indicators and applied the Fuzzy AHP approach to measure the logistic performance index of the member OECD countries [89]. Similarly, Shaygan and Testik [90] collected data from fourteen experts and developed the prioritization model to select

Table 6
TFN conversion scale [44].

"Linguistic scale"	"Triangular Fuzzy scale"	"Triangular Fuzzy reciprocal scale"
"Just equal (JE)"	1,1,1	1, 1, 1
"Equally important (EI)"	1/2, 1, 3/2	0.6, 1, 2
"Weakly important (WI)"	1,3/2,2	0.5, 0.6, 1
"Strongly more important (SMI)"	3/2, 2, 5/2	0.4, 0.5, 0.6
"Very strongly more important (VSMI)"	2,5/2,3	0.3, 0.4, 0.5
"Absolutely more important (AMI)"	5/2, 3, 7/2	0.2, 0.3, 0.4

the information systems for hospitals. The prioritization model provides a roadmap to improve the selection process of the hospital information systems based on the available resources [90]. In this study, the data collected from 23 participants, which is a justifiable data sample in comparison to the above recently published studies and the subjectivity of the Fuzzy AHP approach.

The data collected during the Fuzzy AHP survey are used for the pairwise comparison purpose by converting it into the TFN numbers using geometric mean (GM):

$$GM = [(z_1)(z_2)(z_3) \dots (z_n)]^{1/n}$$

z = each judgment individual weight

n = total number of judgments

The pairwise comparison matrices for the success factors and their categories are developed using the triangular Fuzzy conversion scale given in Table 6.

Step-3: The consistency check for each pairwise matrix is performed by following the criteria defined in Section 7 (step-5). For example, use Eq. (15) to calculate the largest eigenvalue (λ_{max}) for the categories of the reported success factors. In that case, use Eq. (12) to defuzzified the TFN number of the pairwise comparison matrix to a crisp number and develop its Fuzzy crisp matrix (Table 7).

In the next phase, the largest eigenvalue (λ_{max}) is calculated by taking the column sum in Table 7 and then dividing each element of the matrix by the respective column sum. For example, the column sum value of the *organizational management* category is 2.6 and its column value is 1 (Table 7), therefore its normalized value is $1/2.6 = 0.384$ (Table 8). Similarly, the priority weight for each element of the matrix is calculated by taking the average of its respective row. The calculated priority weight of the *organizational management* category is (0.366), which is the average value of its respective row (Table 8).

$$\lambda_{max} = \Sigma([\Sigma C_j] \times W) \quad (15)$$

where ΣC_j = column sum of Matrix $[C]$ (Table 7),

W = priority weight (Table 8), therefore $\lambda_{max} = 2.6 * 0.366 + 6.6 * 0.173 + 3.5 * 0.292 + 5.5 * 0.168 = 4.044$

Based on the value of λ_{max} , we calculated the values of Consistency Index (CI) and Consistency Ratio (CR) as discussed in step-5 of Section 7.

$$CI = \frac{\lambda_{max} - n}{n - 1} = \frac{4.044 - 4}{4 - 1} = 0.014$$

where $n = 4$ is the total number of elements in the matrix (Table 7) and the RI value calculated for $n = 4$ from Table 5.

$$CR = \frac{CI}{RI} = \frac{0.014}{0.9} = 0.015$$

The calculated consistency ratio ($CR = 0.015$) is less than 0.10, therefore the comparison matrix and priority weight chart developed for the give categories is consistent (Table 10, Fig. 5). Moreover, the same approach is used to evaluate the consistency ratio of all the factors of the categories and the results are respectively provided for *organizational management*, *team*, *technology* and *process* categories in Tables 11–14.

Table 7
Fuzzy Crisp Matrix for the categories of success factors.

	Organizational management	Team	Technology	Process
Organizational management	1	2.5	1.5	1.5
Team	0.4	1	0.4	1.5
Technology	0.6	2.5	1	1.5
Process	0.6	0.6	0.6	1
Column Sum	2.6	6.6	3.5	5.5

Table 8
Normalized matrix for the success factor categories.

	Organizational management	Team	Technology	Process	Priority weight
Organizational management	0.384	0.378	0.428	0.272	0.366
Team	0.153	0.151	0.114	0.272	0.173
Technology	0.230	0.378	0.285	0.272	0.292
Process	0.230	0.090	0.171	0.181	0.168

Table 9
Degree of possibility.

	SF3	SF4	SF23	SF21	d (Priority Weight)
Y(SF3 ≥ ...)		1	1	1	1
Y(SF4 ≥ ...)	0.243		0.382	1	0.243
Y(SF23 ≥ ...)	0.841	1		1	0.841
Y (SF21 ≥ ...)	0.217	0.854	0.337		0.217

Step-4: Local weight (LW) for the factors with respect to their categories is calculated using Eq. (2). We first determined the synthesis extent value for the four factors of the *technology* category and use Eq. (2) to calculate the priority weight for each factor. Here the example is just provided for the success factor of the single category i.e., *technology*.

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m x_{gi}^j &= (1, 1, 1) + (2, 2.5, 3) + \dots + (0.5, 0.6, 1) \\ &+ (1, 1, 1) = (14.6, 18.2, 23.0) \\ \left[\sum_{i=1}^n \sum_{j=1}^m x_{gi}^j \right]^{-1} &= \left(\frac{1}{23}, \frac{1}{18.2}, \frac{1}{14.6} \right) = (0.043, 0.054, 0.068) \\ \sum_{i=1}^n \sum_{j=1}^m x_{g1}^j &= (1, 1, 1) + (2, 2.5, 3) + (1, 1.5, 2) + (1, 1.5, 2) \\ &= (5.0, 6.5, 8.0) \\ \sum_{i=1}^n \sum_{j=1}^m x_{g2}^j &= (0.3, 0.4, 0.5) + (1, 1, 1) + (0.3, 0.4, 0.5) \\ &+ (1, 1.5, 2) = (2.6, 3.3, 4.0) \end{aligned}$$

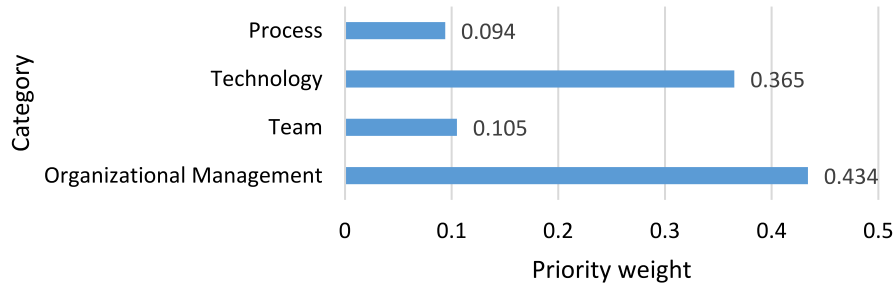


Fig. 5. Priority weight of main categories.

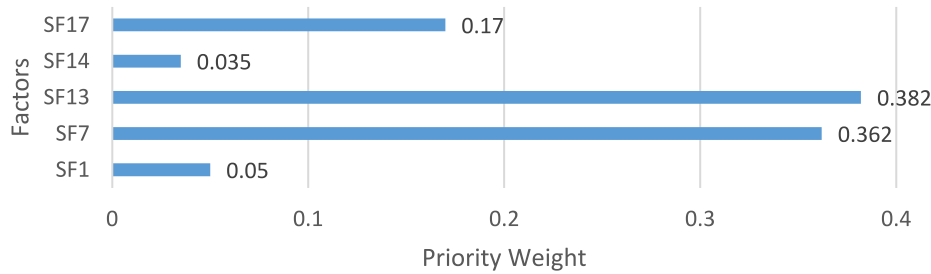


Fig. 6. Priority weight of Organization management factors.

Table 10

Categories pairwise comparison matrix.

	Organizational Management	Team	Technology	Process	Priority Weight
Organizational Management	1,1,1	2, 2.5, 3	1, 1.5, 2	1, 1.5, 2	0.434
Team	0.3, 0.4, 0.5	1,1,1	0.3, 0.4, 0.5	1, 1.5, 2	0.105
Technology	0.5, 0.6, 1	2, 2.5, 3	1,1,1	1, 1.5, 2	0.365
Process	0.5, 0.6, 1	0.5, 0.6, 1	0.5, 0.6, 1	1,1,1	0.094

 $\lambda_{max} = 4.04$, CI = 0.014, CR = 0.015.

$$\sum_{i=1}^n \sum_{j=1}^m X_{g3}^j = (0.5, 0.6, 1) + (2, 2.5, 3) + (1, 1, 1) \\ + (1, 1.5, 2) = (4.5, 5.6, 7.0) \\ \sum_{i=1}^n \sum_{j=1}^m X_{g4}^j = (0.5, 0.6, 1) + (0.5, 0.6, 1) + (0.5, 0.6, 1) \\ + (1, 1, 1) = (2.5, 2.8, 4.0)$$

Following are the synthesis values of four success factors (SF3, SF4, SF23, SF21) of *technology* category that are calculated using Eq. (1):

$$\begin{aligned} SF3 &= (5.0, 6.5, 8.0) \otimes (0.043, 0.054, 0.068) \\ &= (0.215, 0.351, 0.544) \\ SF4 &= (2.6, 3.3, 4.0) \otimes (0.043, 0.054, 0.068) \\ &= (0.111, 0.178, 0.272) \\ SF23 &= (4.5, 5.6, 7.0) \otimes (0.043, 0.054, 0.068) \\ &= (0.193, 0.302, 0.476) \\ SF21 &= (2.5, 2.8, 4.0) \otimes (0.043, 0.054, 0.068) \\ &= (0.107, 0.151, 0.272) \end{aligned}$$

Eq. (7) is used to calculate the degree of possibility as shown in Table 9.

We further used Eq. (9) to calculate the minimum degree of possibility (priority weight) for each of the four *technology* category factors (i.e., SF3, SF4, SF23, SF21).

$$d' (SF3) = \min Y (1, 1, 1) = 1$$

$$d' (SF4) = \min Y (0.243, 0.382, 1) = 0.243$$

$$d' (SF23) = \min Y (0.841, 1, 1) = 0.841$$

$$d' (SF21) = \min Y (0.217, 0.854, 0.337) = 0.217$$

The last column of Table 9 is the degree of possibility (priority local weight) calculated for each success factor of the *technology* category. The final weight vector of all the four factors (SF3, SF4, SF23, SF21) is presented as $W' = (1.00, 0.243, 0.841, 0.217)$. The normalized values of the weight vector are: $W = (0.430, 0.172, 0.238, 0.160)$. The given results revealed that SF3 has the highest normalized priority weight as compared to the other factors (i.e., SF4, SF23, SF21) of the *technology* category.

The detail pairwise comparison results and priority weight chart for the *technology* factors are respectively provided in Table 13 and Fig. 8. The conversion scale given in Table 6 is used to convert the survey data into the TFN and reciprocal Fuzzy numbers. The main purpose of the conversion is to remove the biases and uncertainties in the judgment and evaluation made by the survey participants. Furthermore, the consistency ratio (CR) value calculated for the technology category is $0.07 < 0.1$, which is in the acceptable range (Table 13). It illustrates that based on the above discussion, the opinion and feedback provided by each survey respondent could be considered for generalizing the results.

Similarly, the pairwise comparison matrices and CR value for the success factors of other categories i.e., *organizational management*, *team* and *process* are respectively provided in Tables 11, 12 and 14. We further developed the priority weight chart of each category factors based on the pairwise comparison results. The priority weight charts between the *categories*, *organizational management* category, *team* category, *technology* and *process* category are respectively provided in Figs. 5–9.

Table 11
Organization management factors pairwise comparison matrix.

Cat: (Organizational Management)

	SF1	SF7	SF13	SF14	SF17	Priority Weight
SF1	1,1,1	0.3, 0.4, 0.5	0.4, 0.5, 0.6	1.5, 2, 2.5	0.4, 0.5, 0.6	0.050
SF7	2, 2.5, 3	1,1,1	2, 2.5, 3	0.5, 1, 1.5	1, 1.5, 2	0.362
SF13	1.5, 2, 2.5	0.3, 0.4, 0.5	1,1,1	2, 2.5, 3	2.5, 3, 3.5	0.382
SF14	0.4, 0.5, 0.6	0.6, 1, 2	0.3, 0.4, 0.5	1,1,1	(0.5, 0.6, 1	0.035
SF17	1.5, 2, 2.5	0.5, 0.6, 1	0.2, 0.3, 0.4	1, 1.5, 2	1,1,1	0.170

$\lambda_{max} = 5.4$ CI = 0.1, CR = 0.08.

Table 12
Team factors pairwise comparison matrix.

Cat: (Team)

	SF8	SF6	SF9	SF12	SF15	Priority Weight
SF8	1,1,1	0.4, 0.5, 0.6	1.5, 2, 2.5	0.4, 0.5, 0.6	0.4, 0.5, 0.6	0.108
SF6	1.5, 2, 2.5	1,1,1	2, 2.5, 3	0.5, 1, 1.5	1, 1.5, 2	0.300
SF9	0.4, 0.5, 0.6	0.3, 0.4, 0.5	1,1,1	1.5, 2, 2.5	2, 2.5, 3	0.226
SF12	1.5, 2, 2.5	0.6, 1, 2	0.4, 0.5, 0.6	1,1,1	0.5, 0.6, 1	0.179
SF15	1.5, 2, 2.5	0.5, 0.6, 1	0.3, 0.4, 0.5	1, 1.5, 2	1,1,1	0.187

$\lambda_{max} = 5.5$, CI = 0.12, CR = 0.1.

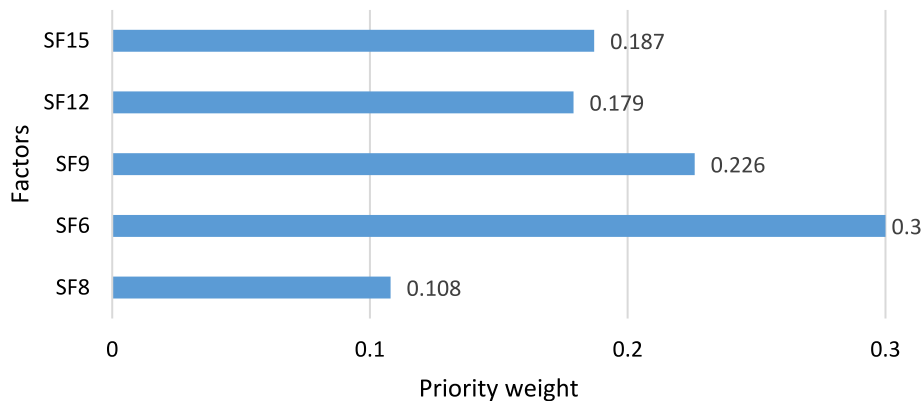


Fig. 7. Priority weight of Team factors.

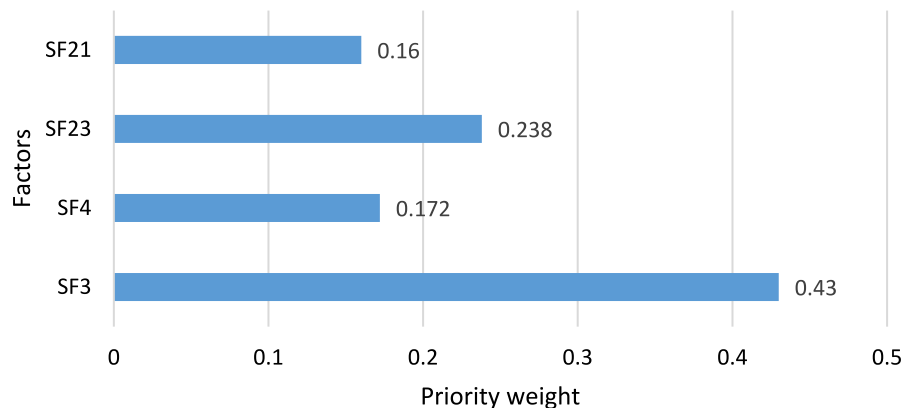


Fig. 8. Priority weight of Technology factors.

Step-5: The local and global weight of the factors and the categories are used to determine their respective prioritization and ranking (Table 15).

Local prioritization or local weight is the significance degree of a specific factor with respect to a specific category. For example, the results given in Table 15 illustrate that the local prioritization of SF3 is higher in comparison to the other factors of technology

category because SF3 has the highest local weight (0.430) value. It demonstrates that the Chinese practitioners consider the significance of SF3 (engineering excellence) for scaling agile method in GSD industry. The main reason might be that it is very important for a specific organization to have excellence in engineering perspectives with respect to both agile and GSD. As in GSD, the team members work across the distributed environment, therefore it is

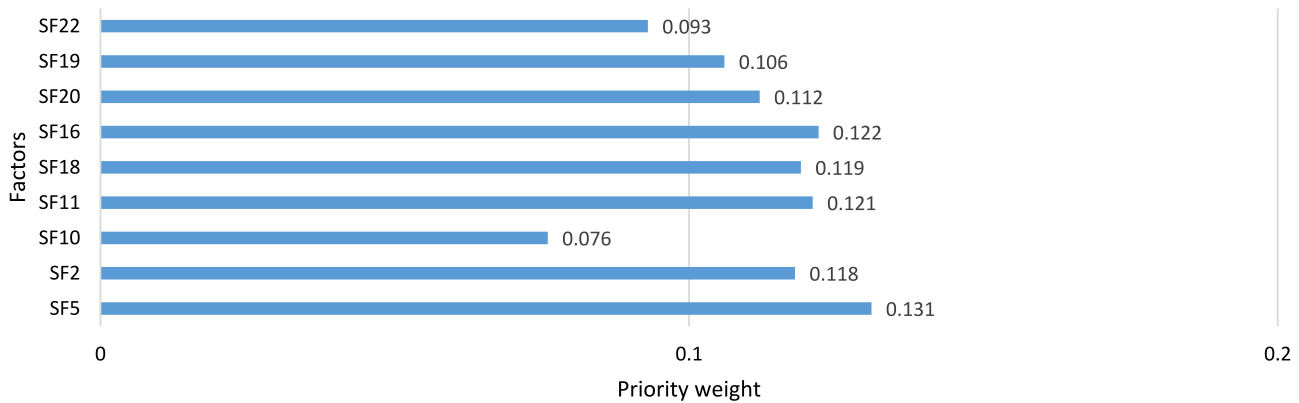


Fig. 9. Priority weight of Process factors.

Table 13

Technology factors pairwise comparison matrix.

Cat: (Technology)

	SF3	SF4	SF23	SF21	Priority Weight
SF3	1, 1, 1	0.4, 0.5, 0.6	0.3, 0.4, 0.5	1.5, 2, 2.5	0.430
SF4	1.5, 2, 2.5	1, 1, 1	0.5, 0.6, 1	0.5, 0.6, 1	0.172
SF23	2, 2.5, 3.0	1, 1.5, 2	1, 1, 1	1, 1.5, 2	0.238
SF21	0.4, 0.5, 0.6	1, 1.5, 2	0.5, 0.6, 1	1, 1, 1	0.160

 $\lambda_{max} = 4.2$, CI = 0.06, CR = 0.07.

quite challenging to scale agile method without excellence in the engineering domain.

Similarly, the global prioritization of the given factors is determined by multiplying each factor local weight with the weight of its respective category (Table 15, Fig. 10). It ranks each factor globally (among all the identified factors) and provides a clear picture of factor significance level. For example, the results given in Table 15 and Fig. 10 shows that SF13 (*communication and control*) is ranked as the most prioritize factor globally (among all the identified factors). The results determine that the survey participants significantly consider the close communication and control of activities while scaling agile method in a globally distributed environment. Communication is a challenging factor in a distributed environment and it is also one of the key aspects of agile methods, therefore proper communication and control channels could positively impact the agile scaling process. Similarly, the global prioritization and ranking of all the factors and categories are determined (Table 15, Fig. 10).

9. Research questions summary

The key research objective of this study is to develop the prioritization taxonomy of the factors that could impact the agile

scaling process in Chinese GSD industry. Two research questions are developed to achieve the given objective (Section 2). The summary of the research questions findings is discussed in the following sections, which is based on the results presented in the previous sections.

9.1. RQ1 Success factors of agile scaling

Snowballing literature survey approach (Section 3.1) is used to address the RQ1. Total 23 factors are identified that could positively impact the agile scaling process in the GSD environment. The factors are further elaborated and systematically discussed in Section 4. Finally, the conceptual mapping of the identified factors is performed by mapping them into the core four categories. Moreover, the literature findings are empirically validated using the survey questionnaire approach to know the perceptions of the Chinese agile process practitioners regarding the reported factors and their conceptual mapping. Findings of the empirical study revealed that majority of the practitioners positively considered the identified factors and they specified them in the domain of Chinese GSD industry. They were further agree with the conceptual mapping of the factors (Section 5).

9.2. RQ2 Success factors taxonomy

The Fuzzy AHP approach is used to develop the prioritization taxonomy of the factors based on their conceptual mapping. The data collected during the second survey study were finally analyzed using the Fuzzy AHP method and assign the prioritization weight to each factor and their respective categories. The implementation details of Fuzzy AHP approach are provided in Section 8 and the summarize prioritization findings are presented as a taxonomy in Figure-11. The given taxonomy (Fig. 11)

Table 14

Process factors pairwise comparison matrix.

Cat: (Process)

	SF5	SF2	SF10	SF11	SF18	SF16	SF20	SF19	SF22	Priority Weight
SF5	1, 1, 1	1, 1.5, 2	2.5, 3, 3.5	0.6, 1, 2	1.5, 2, 2.5	1, 1.5, 2	0.5, 0.6, 1	0.3, 0.4, 0.5	1, 1.5, 2	0.131
SF2	0.5, 0.6, 1	1, 1, 1	0.5, 0.6, 1	1, 1.5, 2	0.4, 0.5, 0.6	1, 1.5, 2	2, 2.5, 3	1, 1.5, 2	1, 1.5, 2	0.118
SF10	0.2, 0.3, 0.4	1, 1.5, 2	1, 1, 1	0.5, 1, 1.5	0.5, 0.6, 1	0.5, 0.6, 1	1, 1.5, 2	0.5, 0.6, 1	0.4, 0.5, 0.6	0.076
SF11	0.5, 1, 1.5	0.5, 0.6, 1	0.6, 1, 2	1, 1, 1	0.2, 0.3, 0.4	2, 2.5, 3	0.5, 1, 1.5	2, 2.5, 3	1, 1.5, 2	0.121
SF18	0.4, 0.5, 0.6	1.5, 2, 2.5	1, 1.5, 2	2.5, 3, 3.5	1, 1, 1	0.4, 0.5, 0.6	0.2, 0.3, 0.4	1, 1.5, 2	0.5, 1, 1.5	0.119
SF16	0.5, 0.6, 1	0.5, 0.6, 1	1, 1.5, 2	0.3, 0.4, 0.5	1.5, 2, 2.5	1, 1, 1	0.4, 0.5, 0.6	2, 2.5, 3	2, 2.5, 3	0.122
SF20	1, 1.5, 2	0.3, 0.4, 0.5	0.5, 0.6, 1	0.6, 1, 2	2.5, 3, 3.5	1.5, 2, 2.5	1, 1, 1	0.4, 0.5, 0.6	0.5, 0.6, 1	0.112
SF19	2, 2.5, 3	0.5, 0.6, 1	1, 1.5, 2	0.3, 0.4, 0.5	0.5, 0.6, 1	0.3, 0.4, 0.5	1.5, 2, 2.5	1, 1, 1	0.6, 1, 2	0.106
SF22	0.5, 0.6, 1	0.5, 0.6, 1	1.5, 2, 2.5	0.5, 0.6, 1	0.6, 1, 2	0.3, 0.4, 0.5	1, 1.5, 2	0.5, 1, 1.5	1, 1, 1	0.093

 $\lambda_{max} = 9.9$, CI = 0.11, CR = 0.07.

Table 15
Local and global prioritization of the identified factors and categories.

Category	Category Weight	Success factor	Local weight (LW)	Local rank (LR)	Global weight (GW)	Global rank (GR)
Organizational Management	0.434	SF1	0.050	4	0.0217	10
		SF7	0.362	2	0.1560	2
		SF13	0.382	1	0.1649	1
		SF14	0.035	5	0.0151	13
		SF17	0.170	3	0.0737	5
Team	0.105	SF8	0.108	5	0.0113	16
		SF6	0.300	1	0.0315	8
		SF9	0.226	2	0.0237	9
		SF12	0.179	4	0.0187	12
		SF15	0.187	3	0.0196	11
Technology	0.365	SF3	0.430	1	0.1569	3
		SF4	0.172	3	0.0627	6
		SF23	0.238	2	0.0868	4
		SF21	0.160	4	0.0584	7
Process	0.094	SF5	0.131	1	0.0123	14
		SF2	0.118	5	0.0110	19
		SF10	0.076	9	0.0071	23
		SF11	0.121	3	0.0113	17
		SF18	0.119	4	0.0111	18
		SF16	0.122	2	0.0114	15
		SF20	0.112	6	0.0105	20
		SF19	0.106	7	0.0099	21
		SF22	0.093	8	0.0087	22

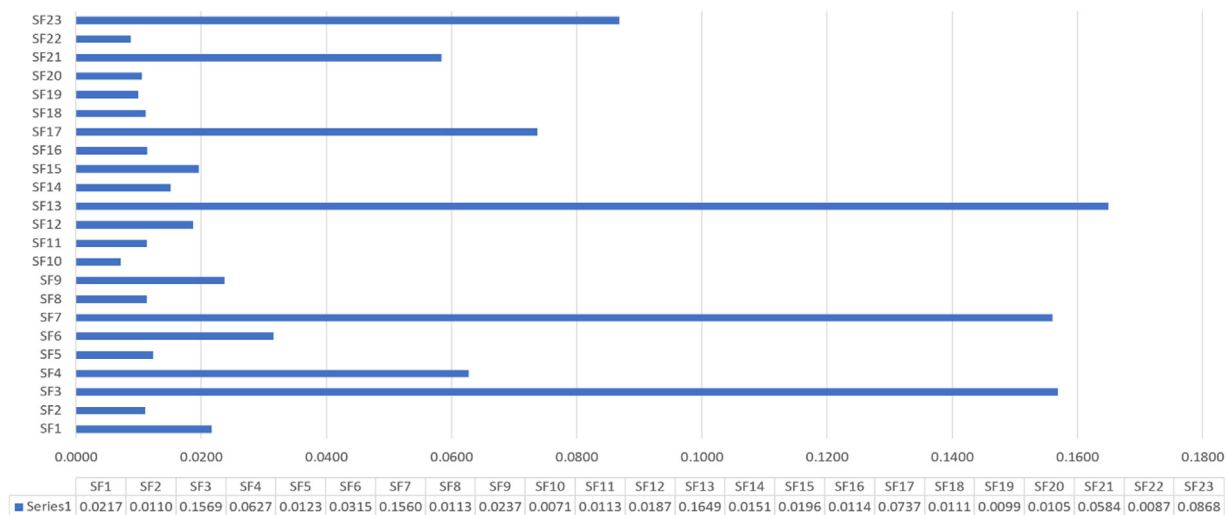


Fig. 10. Global weight of identified factors.

provides the significance weight of the key factors and the categories that could impact the scaling process of agile practices. The results revealed that *organizational management* (0.434) is the most prioritize category because its prioritization weight is greater than the other three categories. Therefore, *organizational management* is considered as the most significant category of factors for scaling agile methods and it is important that Chinese GSD organizations focus on the factors of the *organizational management* category at first priority (Fig. 11). The second most common category is *technology* (0.365) and its factors mainly focus on the technological aspects of the agile scaling process.

The given taxonomy also shows the prioritization weight for each identified factor (Fig. 11). Prioritization weight values given in Fig. 11 are based on the results reported in Table 15, however, we round the global weight to only three decimal places. It makes the taxonomy results more clear and understandable.

The success factor, SF13 (*communication and Control*, LW 0.382, GW:0.165) has the highest priority weight as compared to all the other factors locally and globally. It illustrates that the Chinese GSD industry practitioners consider SF13 both locally (same

category) and globally (in comparison to the factors of other categories) the most significant factor for scaling the agile methods. The other common and priorities factors are SF7 (*arrange social activities*, LW: 0.362, GW: 0.157), SF3 (*engineering excellence*, LW:0.430, GW:0.157) and SF23 (*maturity model for agile transformation*, LW: 0.238, GW:0.086). These are the key areas where the GSD organizations should strongly focus in order to scale the agile methods.

10. Threats to validity

In this section, we discuss the following validity threats [91] that are associated with different activities performed in this study:

10.1. Internal validity threats

The literature survey data are collected using the informal snowballing review approach and there is a chance of missing

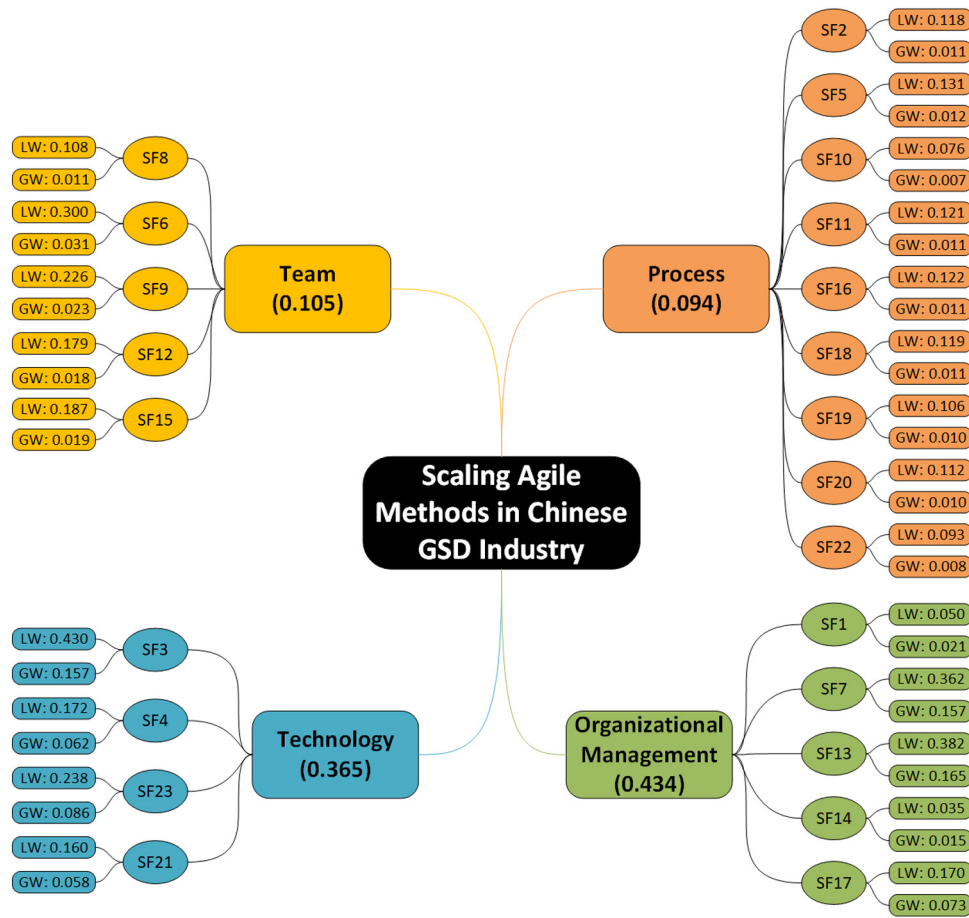


Fig. 11. Prioritization taxonomy of identified success factors.

the relevant data. It might be a threat to the internal validity of the study findings. We tried to eliminate this threat by using the study keywords and exploring the Google Scholar search engine for relevant studies. Moreover, the given threat is minimized by participating all the authors in the review process and confirm the review findings during the consensus meetings.

10.2. External validity threats

The external validity threats refer to the degree to which the study findings could be generalize. The empirical data are only collected from 83 survey participants, because of lack of resources and time to physically approach the targeted population. The given data sample might be small to generalize the study results, however, we plan to extend this study with collaboration from different other research groups and organizations. Finally, it is worth mentioning that studies conducted in same sample size range are regarded as valuable contribution in qualitative software engineering research community. For example, [72] conducted industrial survey study and assessed the research hypotheses based on the data collected from 54 participants. Similarly, [92] and [51] respectively collected data from 81 and 35 respondents.

10.3. Construct validity threats

Construct validity in survey research refers to the extent to which survey questionnaire actually measures the targeted variables. In this study, the questionnaire is developed based on the identified factors (variables), which is further evaluated by conducting the pilot study with the qualitative software engineering

research experts (Section 3.2.1). Moreover, the survey constructs are empirically interpreted based on the data collected from the industrial practitioners. The survey results revealed that most of the respondents agree to consider the identified factors for the agile scaling process.

10.4. Threats to statistical conclusions

There is a possible threat to the validity of statistical conclusions, because the data were statistically analyzed and generalized. In order to minimize this threat, we followed the guidelines provided by [50] to select the most appropriate statistical technique based on the type of the data. Moreover, the pilot evaluation study was conducted with the software engineering experts to ensure the structure of the survey instrument, sampling procedure and survey assessment scale.

11. Implications and future directions

This study provides a thorough overview of the agile scaling process in the domain of Chinese GSD industry. The available literature is investigated to know the key dimensions of agile methods and the empirical study revealed the significance of the reported success factors and their categories based on the Fuzzy AHP prioritization. The literature findings and the empirical evaluation improve the body of knowledge for academic research. On the other hand, the prioritization taxonomy of the factors contributes to the industry by providing a robust framework that offers the guideline for scaling agile methods in Chinese GSD industry. It will enhance the understanding of the practitioners to

Table A.16

Survey questionnaire sample.

Section 1: Participant demographic details

Name (optional)				
Gender	Male	Female		
Job title				
Organization name				
Organization location (city)				
Please specify the size of your organizations.	Small	Medium	Large	
Which agile framework are you practicing?	Extreme Programming (XP)	Scrum	Lean	Kanban
Other (write name)				
Total Agile/GSD working experience (years)				

Section 2: Success factors and their categories

The list of the identified 23 success factors and their respective categories are provided in this section with the aim to validate based on the experience and understanding of the survey participants. Please record your feedback against the given five-point Likert scale (SD: strongly disagree; D: disagree; N: neutral; A: agree; SA: strongly agree) by marking the appropriate box.

For example, if you consider that the success factor (SF), *leadership support* is positively significant factor for the agile scaling process, then tick mark either agree (A) or strongly agree (SA) based on your knowledge and understanding. Follow the same assessment approach for all the other factors and their categories.

Factors and categories	SD	D	N	A	SA
Organizational management category					
Leadership support (SF1)					
Arrange social activities (SF7)					
Communication and control (SF13)					
Organizational vision (SF14)					
Skilled human resources (SF17)					
Please provide any additional factors or suggestions regarding the <i>organizational management</i> category:					

Team category	SD	D	N	A	SA
Customer involvement (SF8)					
Agile champions and change agents (SF6)					
Creating self-organizing agile teams (SF9)					
Training and coaching (SF12)					
Continuous learning environment (SF15)					
Please provide any additional factors or suggestions regarding the <i>team</i> category:					
Technology category					
Engineering excellence (SF3)					
Tools and infrastructure (SF4)					
Maturity model for agile transformation (SF23)					
Use a large-scale agile framework (SF21)					
Please provide any additional factors or suggestions regarding the <i>technology</i> category:					

Process category	SD	D	N	A	SA
Communities of practices (SF5)					
Knowledge acquisition (SF2)					
Selecting and customizing the agile approach (SF10)					
Workflow transparency (SF11)					
Start with minimum viable product (MVP) (SF18)					
Instant feedback (SF16)					
Build a collaborative culture (SF20)					
Create a single product backlog (SF19)					
Planning for scaling (SF22)					
Please provide any additional factors or suggestions regarding the <i>process</i> category:					

consider the most significant dimensions of agile methods before initiating the scaling process.

The domain of this study is Chinese GSD industry and in future, we plan to extend and evaluate the findings of this study

Table B.17

Details of the survey respondents.

S. No	Gender	Job title	City	Organization size	Working experience(years)	Agile approach
1.	Male	Agile coach	Nanjing	Medium	5	XP
2.	Male	Product owner	Nanjing	Small	7	XP
3.	Male	Scrum master	Nanjing	Small	4	Scrum
4.	Male	Senior developer	Nanjing	Small	8	Scrum
5.	Male	Agile consultant	Beijing	Medium	12	Scrum/XP
6.	Female	Designer	Nanjing	Small	3	Kanban
7.	Male	Scrum master	Shanghai	Large	6	Lean and Kanban
8.	Female	Tester	Shanghai	Small	3	Crystal
9.	Male	Developer	Nanjing	Medium	4	XP
10.	Male	Program manager	Shenzhen	Medium	9	Scrum
11.	Male	Product owner	Shenzhen	Large	8	XP/Lean
12.	Male	Developer	Hangzhou	Small	5	Kanban
13.	Male	Agile coach	Nanjing	Medium	7	Scrum
14.	Male	Analyst	Hangzhou	Large	5	Kanban
15.	Male	System administrator	Hangzhou	Small	8	Scrum
16.	Female	Developer	Nanjing	Large	5	XP
17.	Male	Scrum master	Nanjing	Medium	5	Scrum
18.	Male	Software architect	Beijing	Small	6	Feature-Driven Development (FDD)/Scrum
19.	Male	Agile consultant	Suzhou	Medium	11	Scrum/Lean/Kanban
20.	Female	Scrum master	Hangzhou	Large	7	Scrum
21.	Male	Tester	Guangzhou	Medium	5	Crystal
22.	Male	Analyst	Guangzhou	Medium	7	Scrum
23.	Male	Architect	Nanjing	Small	6	XP
24.	Male	Proxy-PO	Nanjing	Medium	8	Lean
25.	Male	Agile program manager	Beijing	Medium	10	Scrum/Kanban
26.	Female	Junior developer	Hefei	Small	3	XP
27.	Male	Scrum master	Chongqing	Large	11	Scrum
28.	Male	Consultant	Chengdu	Large	9	Scrum/XP
29.	Male	Designer	Nanjing	Small	4	Kanban
30.	Male	Agile coach	Shanghai	Large	8	Scrum/Lean
31.	Female	System administrator	Suzhou	Small	3	XP
32.	Male	Senior software developer	Shenzhen	Medium	7	Lean
33.	Male	Product owner	Shanghai	Large	8	XP
34.	Male	Developer	Beijing	Medium	5	Kanban
35.	Male	Scrum master	Guangzhou	Medium	6	Scrum
36.	Male	Agile consultant	Suzhou	Small	10	Scrum/XP/Kanban/Lean
37.	Male	Tester	Chongqing	Small	5	Lean
38.	Male	Software designer	Nanjing	Medium	3	FDD
39.	Male	Program manager	Shenzhen	Large	9	Scrum/Lean/XP
40.	Male	Team leader	Shanghai	Large	9	XP
41.	Female	System analyst	Shenzhen	Small	6	Lean
42.	Male	Quality engineer	Hangzhou	Small	4	Kanban
43.	Male	Software developer	Suzhou	Medium	6	Crystal/XP
44.	Male	Scrum master	Chongqing	Medium	8	Scrum/XP
45.	Male	Team manager	Xian	Large	5	Lean
46.	Male	Agile coach	Nanjing	Small	7	Scrum
47.	Male	Product owner	Chengdu	Small	10	Scrum
48.	Female	Consultant	Suzhou	Medium	5	Lean
49.	Male	System administrator	Hefei	Large	4	Kanban
50.	Male	Developer	Guangzhou	Medium	6	XP
51.	Male	Agile coach	Shanghai	Large	8	Lean/FDD
52.	Male	Product owner	Hangzhou	Large	10	Scrum/XP
53.	Female	Product owner	Nanjing	Small	7	Lean/Kanban
54.	Female	System analyst	Beijing	Medium	4	XP
55.	Male	Tester	Shenzhen	Small	5	Kanban
56.	Male	Quality engineer	Hangzhou	Large	5	Scrum
57.	Male	Scrum master	Chongqing	Medium	8	Scrum
58.	Female	Senior software developer	Wuhan	Small	6	XP
59.	Male	Agile coach	Chengdu	Large	9	Scrum
60.	Male	Designer	Wuhan	Small	4	Kanban
61.	Male	Tester	Nanjing	Medium	7	Lean
62.	Male	Manager	Nanjing	Small	6	Kanban
63.	Female	Product owner	Beijing	Small	8	XP/Lean/Kanban
64.	Male	Agile consultant	Shanghai	Medium	5	Scrum
65.	Male	Quality engineer	Suzhou	Medium	8	Lean
66.	Male	Scrum master	Hangzhou	Large	8	XP
67.	Female	Developer	Shenzhen	Medium	4	Kanban
68.	Male	Agile coach	Wuhan	Medium	7	Scrum/Lean
69.	Male	Product owner	Nanjing	Small	10	XP/Kanban/Scrum
70.	Female	Team lead	Shanghai	Medium	6	XP
71.	Male	System administrator	Beijing	Large	3	Crystal/FDD
72.	Male	Tester	Beijing	Medium	8	Scrum
73.	Male	Agile coach	Nanjing	Large	11	Kanban
74.	Male	Scrum master	Wuhan	Small	7	Lean/Scrum
75.	Male	Program manager	Hangzhou	Large	9	Scrum

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Table B.17 (continued).

S. No	Gender	Job title	City	Organization size	Working experience(years)	Agile approach
76.	Female	System analyst	Xian	Medium	3	Kanban/XP
77.	Male	Product owner	Wuhan	Large	6	Lean
78.	Female	Analyst	Nanjing	Medium	4	XP
79.	Male	Team lead	Nanjing	Small	7	Scrum
80.	Male	Manager	Shanghai	Small	9	XP/Lean
81.	Male	Consultant	Suzhou	Medium	5	Scrum
82.	Female	Senior software developer	Shenzhen	Large	8	Lean
83.	Male	Product owner	Xian	Small	5	XP

Table C.18

Survey questionnaire for Fuzzy AHP data.

Section-A (Respondent Information)

Name (optional)	Job position
Agile process or global software development (GSD) work experience (Years)?	
Email Address	
Current organization name and address in China	
Have you ever participated in agile transformation project?	Yes No

Section-A1 (Organization Detail)

Current organization business domains? (You may tick more than one)	Global/offshore Development	Collocated/single site development
	Agile software development	Other
Organization size	Small	Medium
	Large	Not sure
Type of your current working organization	National	Multinational
	Not sure	Other

Does your organization use agile transformation maturity model (Please specify)?

For how many years the organization use agile transformation standard/model?

Section-B (List of identified success factors and their respective categories)

Category	Success Factors
	Leadership support (SF1)
	Arrange social activities (SF7)
Organizational management	Communication and control (SF13)
	Organizational vision (SF14)
	Skilled human resources (SF17)
	Customer involvement (SF8)
	Agile champions and change agents (SF6)
Team	Creating self-organizing agile teams (SF9)
	Training and coaching (SF12)
	Continuous learning environment (SF15)
	Engineering excellence (SF3)
Technology	Tools and infrastructure (SF4)
	Maturity model for agile transformation (SF23)
	Use a large-scale agile framework (SF21)
	Communities of practices (SF5)
	Knowledge acquisition (SF2)
	Selecting and customizing the agile approach (SF10)
	Workflow transparency (SF11)
Process	Start with minimum viable product (MVP) (SF18)
	Instant feedback (SF16)
	Build a collaborative culture (SF20)
	Create a single product backlog (SF19)
	Planning for scaling (SF22)

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in different other countries and continents to conduct region-wise comparative study for agile scaling process. It will help to identify the most common challenges and process areas between

the Chinese and GSD industries in different other regions. The findings of the comparative study will further be used to develop an agile scaling and management maturity model. The model

Table C.18 (continued).

Pairwise compare the listed success factors and their respective categories using the given evaluation scale. For example, based on your preference, if a factor at the left side of the table has more significance in comparison to the right side, then put the check mark (✓) sign at the left side of the *just equal (JE)* scale (Section-B1). Similarly, compare the categories of the factors using the same evaluation approach (Section-B1). Suppose that the category at the right side is more significant from the one at the left side then provide the check mark on the right side of the *just equal (JE)* scale. Section-B1 is the evaluation hierarchical structure of the success factors and their categories in order to develop the pairwise comparison matrices and determine the priority weight of each variable. For example, what would be the priority weight of SF1 in comparison to SF2 etc.

Description	Significance intensity
"Just equal (JQ)"	1,1,1
"Equally important (Elp)"	0.5,1,1.5
"Weakly important (Wlp)"	1,1.5,2
"Strong more important (SMlp)"	1.5,2,2.5
"Very strong more important (VSMlp)"	2,2.5,3
"Absolutely more important (AMlp)"	2.5, 3,3.5

Section-B1 (Success factors and categories pairwise comparison)**Organization management success factors**

Alternative	AMlp	VSMlp	SMlp	Wlp	El	JQ	Elp	Wlp	SMlp	VSMlp	AMlp	Alternative
SF1												SF1
SF1												SF7
SF1												SF13
SF1												SF14
SF1												SF17
SF7												SF7
SF7												SF13
SF7												SF14
SF7												SF17
SF13												SF13
SF13												SF14
SF13												SF17
SF14												SF14
SF14												SF17
SF17												SF17

Team success factors

Alternative	AMlp	VSMlp	SMlp	Wlp	Elp	JQ	Elp	Wlp	SMlp	VSMlp	AMlp	Alternative
SF8												SF8
SF8												SF6
SF8												SF9
SF8												SF12
SF8												SF15
SF6												SF6
SF6												SF9
SF6												SF12
SF6												SF15
SF9												SF9
SF9												SF12
SF9												SF15
SF12												SF12
SF12												SF115
SF15												SF15

Technology success factors

Alternative	AMlp	VSMlp	SMlp	Wlp	Elp	JQ	Elp	Wlp	SMlp	VSMlp	AMlp	Alternative
SF3												SF3
SF3												SF4
SF3												SF23
SF3												SF21

(continued on next page)

Table C.18 (continued).

SF4												SF4
SF4												SF23
SF4												SF21
SF23												SF23
SF23												SF21
SF21												SF21
Process success factors												
Alternative	AMIp	VSMIp	SMIp	WIp	EIp	JQ	EIp	WIp	SMIp	VSMIp	AMIp	Alternative
SF5												SF5
SF5												SF2
SF5												SF10
SF5												SF11
SF5												SF18
SF5												SF16
SF5												SF20
SF5												SF19
SF5												SF22
SF2												SF2
SF2												SF10
SF2												SF11
SF2												SF18
SF2												SF16
SF2												SF20
SF2												SF19
SF2												SF22
SF10												SF10
SF10												SF11
SF10												SF18
SF10												SF16
SF10												SF20
SF10												SF19
SF10												SF22
SF11												SF11
SF11												SF18
SF11												SF16
SF11												SF20
SF11												SF19
SF11												SF22
SF18												SF18
SF18												SF16
SF18												SF20
SF18												SF19
SF18												SF22
SF16												SF16
SF16												SF20
SF16												SF19
SF16												SF22
SF20												SF20
SF20												SF19
SF20												SF22
SF19												SF19
SF19												SF22
SF22												SF22

(continued on next page)

Table C.18 (continued).

Pairwise evaluation of the Categories Organizational Management, Technology, Team, Process												
Alternative	AMIp	VSMIp	SMIp	WIp	Elp	JQ	Elp	WIp	SMIp	VSMIp	AMIp	Alternative
Organizational Management												Organizational Management
Organizational Management												Technology
Organizational Management												Team
Organizational Management												Process
Technology												Technology
Technology												Technology
Technology												Process
Team												Team
Team												Process
Process												Process

will evaluate the agile scaling capabilities of an organization and provide best practices for future improvements.

12. Conclusions

This study is conducted with the aim to develop the prioritization taxonomy of the factors that could positively impact the agile scaling process activities in the domain of GSD. The available literature studies are explored to identify the key factors that are significant for the agile scaling process. The literature survey is ended by identifying total 23 success factors, that are further conceptually classified across four core categories i.e., *organizational management, team, technology and process*. The empirical data are collected from 83 practitioners working in the domain of Chinese GSD industry to validate the literature findings and empirically assess the conceptual mapping of the identified factors. The Fuzzy AHP approach is used to prioritize the empirically validated success factors and their categories based on the expert's opinion. The Fuzzy AHP pairwise comparison results revealed that *organizational management* is the most prioritize category recommend by the experts. Moreover, *SF13 (communication and Control)* is considered the most significant factor that could positively impact the agile methods in the Chinese GSD industry. In conclusion, the prioritization weight determined for the factors and their respective categories provide a taxonomy that offers recommendations to the organizations who are willing to move from traditional agile environment to globally distributed. The prioritize factors and their categorization are the key process areas that need significant attention and consideration while managing agile software development activities in Chinese GSD organizations.

CRedit authorship contribution statement

Arif Ali Khan: Conceptualization, Approach Targeted Population, Data collection, Data analysis, Writing - original draft. **Mohammad Shameem:** Review literature findings, Conduct empirical data analysis. **Mohammad Nadeem:** Review literature findings, Conduct empirical data analysis. **Muhammad Azeem Akbar:** Review literature and Empirical findings, Address reviewers' comments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Survey questionnaire sample (see [Table A.16](#)).

Appendix B

Details of the survey respondents (see [Table B.17](#)).

Appendix C

Survey questionnaire for Fuzzy AHP data (see [Table C.18](#)).

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